

Multi-Resolution Global Horizontal Irradiance Forecasting at Electric-Vehicle Manufacturing Locations: TimesNet and Dilated Recurrent Networks

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Abstract

This journal extension investigates global horizontal irradiance forecasting at ten locations associated with electric-vehicle factories across complementary temporal resolutions. It consolidates an hourly TimesNet study and a monthly DilatedRNN study under a common methodological description and defines the additional daily and matched-horizon experiments required for a fair cross-model comparison. The hourly study uses 336 hours of context to predict the following 72 hours, while the monthly study predicts one month ahead from short multiyear series. Because their current targets, horizons, sample sizes, and primary metrics differ, their published errors are reported as prior evidence rather than compared directly. The completed extension will evaluate TimesNet and DilatedRNN under identical chronological splits, forecast origins, post-processing rules, and location-level aggregation at every shared temporal resolution. [Draft note: Replace this final sentence with consolidated numerical results only after the missing matched experiments have been executed.]

Keywords: global horizontal irradiance, solar forecasting, TimesNet,

1. Introduction

The rapid expansion of photovoltaic generation has increased the importance of solar-resource forecasts for system planning, energy management, and grid integration. Forecast usefulness is nevertheless conditional on temporal resolution, prediction horizon, available covariates, and evaluation protocol [1, 2, 3]. A model that performs well for an hourly three-day trajectory does not necessarily remain competitive for daily or monthly planning.

Industrial decarbonization makes this distinction especially relevant for manufacturing facilities. Electric-vehicle factories provide a geographically diverse case study because their locations span different latitudes, seasonal cycles, and atmospheric regimes. In this work, the factories identify the geographic points at which solar irradiance is studied; no production, consumption, or proprietary operational data are used.

Two preceding studies motivate this journal extension. The first configured a compact TimesNet for direct 72-hour global horizontal irradiance (GHI) forecasting and evaluated the 24-, 48-, and 72-hour prefixes. The second evaluated a Dilated Recurrent Neural Network (DilatedRNN) for one-month-ahead forecasting from short monthly series. Their results cannot yet establish which architecture is preferable across scales because the targets, context lengths, forecast origins, competitors, and primary metrics differ.

This extension therefore has three objectives: (i) provide a unified and reproducible description of the data, transformations, architectures, and

evaluation rules; (ii) evaluate TimesNet and DilatedRNN at shared hourly, daily, and longer planning horizons; and (iii) characterize when performance changes with location, horizon, and climate-related variability. Its intended contribution is a matched multi-resolution comparison, rather than a direct juxtaposition of incomparable errors from the preceding manuscripts.

The remainder of the article reviews the theoretical background, documents the data and feature engineering, defines the common experimental protocol, presents the evaluation metrics separately, and analyzes aggregate and adverse cases. Full hyperparameter grids and location-level results are assigned to the appendix.

2. Theoretical Background and Related Work

2.1. Solar irradiance and forecasting horizons

Global horizontal irradiance is the total irradiance incident on a horizontal surface,

$$\text{GHI} = \text{DHI} + \text{DNI} \cos(\theta_z), \quad (1)$$

where DHI and DNI denote diffuse horizontal and direct normal irradiance, and θ_z is the solar zenith angle [4]. The NSRDB provides a satellite-derived, model-based irradiance reference rather than measurements collected at the factories [5].

Hourly forecasts preserve the night–day cycle and short atmospheric variations. Daily aggregation suppresses intraday structure while retaining weather-driven and seasonal changes. Monthly aggregation emphasizes the annual cycle but produces very short training series. Accordingly, forecast

horizon must be stated jointly with temporal resolution; terms such as “short-term” and “long-term” are ambiguous without both quantities.

2.2. Time-series learning for solar forecasting

Solar forecasting uses statistical baselines, tree-based learners, recurrent networks, convolutional models, and attention-based architectures [2, 6]. Transparent persistence and seasonal-naive forecasts remain essential because they reveal whether added architectural complexity improves upon temporal continuity and seasonality.

TimesNet transforms dominant one-dimensional temporal variations into two-dimensional representations and processes within-period and between-period variation with convolutional kernels [7]. This representation is well suited to signals with multiple periodicities, including the diurnal and seasonal components of GHI.

Dilated recurrent networks connect a state to selected earlier states through increasing temporal skips. The shorter computational paths can expose dependencies at several scales without extending a conventional recurrent chain one step at a time [8]. Global models additionally share parameters across related locations while using a site identifier to preserve location-specific behavior [9].

2.3. Industrial transition and climate variability

[Draft note: Expand this subsection with verified primary and high-quality sources on industrial electrification, photovoltaic adoption at manufacturing sites, the energy transition through 2050/2060, and the documented

Caçapava industrial example. Do not state a legal obligation unless the applicable jurisdiction and regulation are identified.]

Cloud variability deforms otherwise regular solar profiles and generally increases forecast difficulty. The final literature review will distinguish irradiance variability, cloud cover, and precipitation instead of treating low GHI alone as proof of a rainy day.

3. Data and Feature Engineering

The study uses the GOES Aggregated Physical Solar Model v4 product from the National Solar Radiation Database. The common geographic sample comprises ten locations associated with electric-vehicle factories in Brazil, Mexico, and the United States. Coordinates select the corresponding NSRDB grid cells and are not interpreted as factory-level irradiance measurements.

Table 1 separates measured or retrieved fields from derived attributes. This distinction is necessary because the hourly and monthly studies currently use different representations. The unified experiments will fit every transformation on pre-test data only and will document the exact input available at each forecast origin.

Before modeling, each series is audited for ordering, duplicate timestamps, coverage, missing values, non-finite values, and physically invalid negative GHI. Aggregation is performed only after this audit. Scaling parameters, quantization boundaries when used, and other learned transformations are estimated from training data and then reused unchanged in validation and test. This ordering prevents information from future periods from influencing model development.

Table 1: Data fields and derived variables used in the unified study.

Group	Variable	Role
Target	GHI	Mean irradiance in W/m^2 at the selected temporal resolution.
Time	Timestamp	Chronological ordering, local-time conversion, and forecast origins.
Location	Site identifier	Categorical identifier for global models.
Geography	Latitude/longitude	NSRDB cell selection and geographic analysis; not a dynamic covariate.
Lagged input	GHI lags	Past target values available before the forecast origin.
Statistics	Moving averages	Trailing summaries fitted without future observations.
Calendar	Cyclical encodings	Sine/cosine representation of hour, day, or month when applicable.
Physics	Solar elevation	Deterministic nighttime post-processing for hourly predictions.

Table 2 reports the grid points already used by the hourly study. A climate-zone column will be added only after assigning every location with a single documented classification source.

Table 2: Factory-associated NSRDB grid points used in the study.

Location	Latitude	Longitude	UTC offset
BMW San Luis Potosí	21.97	−100.86	−6
BYD Camaçari	−12.67	−38.30	−3
Ford Rouge	42.29	−83.18	−5
GM Factory Zero	42.37	−83.06	−5
Hyundai Georgia	32.17	−81.46	−5
Lucid Casa Grande	32.85	−111.78	−7
Rivian Normal	40.49	−89.06	−6
Tesla Fremont	37.49	−121.94	−8
Tesla Nevada	39.53	−119.42	−8
Tesla Texas	30.21	−97.62	−6

4. Methodology

The common pipeline is divided into three blocks following the presentation logic of the reference journal article. Each block has an explicit input, transformation, output, and pseudocode. Figure 1 summarizes their relationship.

4.1. Block A: data preparation

Block A converts audited NSRDB series into resolution-specific supervised examples. All calculations for an origin t use only values available

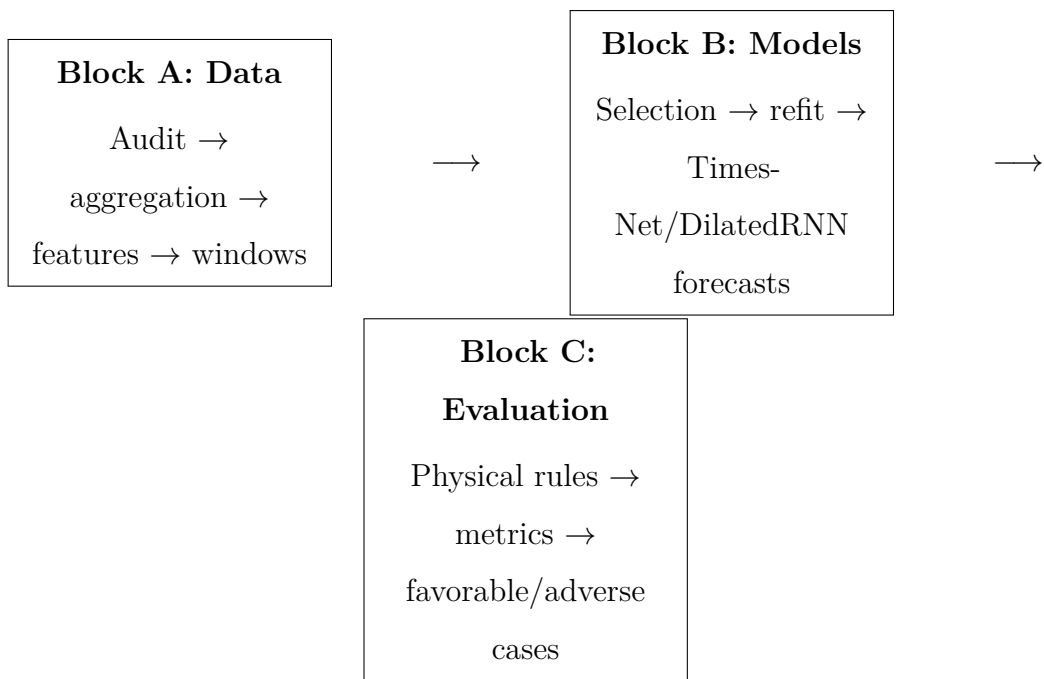


Figure 1: Unified experimental pipeline. Each block is detailed by a separate pseudocode and explanatory subsection.

before or at that origin.

Algorithm 1 Block A—data preparation and temporal windows

Require: Raw series, locations $\mathcal{L}$, resolution r , context c , horizon h

Ensure: Chronological training, validation, and test windows

- 1: **for** each location $\ell \in \mathcal{L}$ **do**
 - 2: audit timestamps, missing values, duplicates, and invalid GHI
 - 3: aggregate GHI at resolution r
 - 4: split the series chronologically before fitting transformations
 - 5: fit scaling and derived-feature rules on training data only
 - 6: create (c, h) windows without crossing partition boundaries
 - 7: **end for**
-

The output of Block A is a set of origin-aligned samples shared by every model within the same experiment. This prevents changes in sample availability from being mistaken for architectural gains.

4.2. Block B: model development and forecasting

TimesNet and DilatedRNN will be trained with the same samples and forecast origins at each shared resolution. Architecture-specific context lengths may be studied as an ablation, but the primary comparison will keep the available information equivalent.

4.3. Block C: physical post-processing and evaluation

The favorable and unfavorable examples are selected after computing all location–origin errors, but the selection rule must be declared before their interpretation. A weather case will be labeled rainy or cloudy only when an independent meteorological field or source confirms that condition.

Algorithm 2 Block B—matched training and forecasting

Require: Training and validation windows, predefined model search spaces

Ensure: Fixed models and raw forecasts for every test origin

- 1: **for** each model $m \in \{\text{TimesNet}, \text{DilatedRNN}, \text{baselines}\}$ **do**
 - 2: select hyperparameters using validation data only
 - 3: reinitialize and refit the selected configuration on pre-test data
 - 4: **for** each chronological test origin **do**
 - 5: predict the complete horizon without future target information
 - 6: **end for**
 - 7: **end for**
-

Algorithm 3 Block C—post-processing, metrics, and case selection

Require: Raw forecasts, reference targets, location metadata

Ensure: Aggregate, local, horizon-wise, and adverse-case results

- 1: invert training-fitted transformations
 - 2: apply the same non-negativity rule to every eligible model
 - 3: apply the documented solar-elevation rule to hourly forecasts
 - 4: compute every metric first by location and then its macro-average
 - 5: rank models under the declared primary metric
 - 6: select favorable and unfavorable cases using a rule fixed in advance
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5. Experimental Setup

The preceding hourly and monthly experiments are retained as evidence from the two source manuscripts. They are not treated as a matched comparison. The extension adds only experiments that align resolution, context, horizon, origins, and evaluation rules for TimesNet and DilatedRNN.

Table 3: Existing evidence and experiments required by the journal extension.

Resolution Model		Context	Horizon	Status
Hourly	TimesNet	336 h	24/48/72 h	Complete source experiment
Hourly	DilatedRNN	To be matched	24/48/72 h	Required
Daily	TimesNet	To be fixed	Multiple days	Required
Daily	DilatedRNN	To be matched	Multiple days	Required
Monthly	DilatedRNN	12 months	1 month	Complete source experiment
Monthly	TimesNet	12 months	1 month	Required
Multi-month	Both	To be matched	To be fixed	Required only if defined as long-term

5.1. Hourly protocol

The available TimesNet protocol uses 336 hourly observations as context and predicts 72 hours directly. The 24- and 48-hour results are prefixes of

the same 72-hour output. Training spans 2019–2022, 2023 is used for model selection, the final model is refitted on 2019–2023, and 2024 is reserved for retrospective testing.

5.2. *Monthly protocol*

The available DilatedRNN protocol uses a 12-month context and predicts the next monthly mean GHI. It evaluates the 12 chronological origins of 2024 after training and model selection on pre-test observations. This is a one-step monthly task; it will not be described as a multi-month forecast.

5.3. *Daily and matched long-horizon protocols*

[Draft note: Execute the daily TimesNet and DilatedRNN experiments at identical origins. Fix the daily horizons before inspecting test performance. If “long-term” means multiple monthly steps, execute both architectures with the same direct or recursive forecast strategy and report each horizon separately.]

The primary configuration table will contain only the parameters needed to understand and reproduce the comparison. Complete search spaces, selected epochs by seed, and secondary settings are assigned to Appendix Appendix A.

6. Evaluation Metrics

Let y_i denote a reference GHI value, $\hat{y}_i$ its forecast, $e_i = y_i - \hat{y}_i$, $\bar{y}$ the mean reference value, and n the number of evaluated targets. Metrics are calculated by location before the unweighted macro-average is formed.

Table 4: Verified configurations in the two source experiments.

Item	Hourly TimesNet	Monthly DilatedRNN
Context/output	336 h / 72 h	12 months / 1 month
Main width	$d_{\text{model}} = 8, d_{\text{ff}} = 16$	Three recurrent layers, 16 units each
Temporal blocks	One TimesBlock, $k = 3$	Dilations 1–2–4
Optimizer	Adam, learning rate 10^{-3}	Adam, learning rate 10^{-3}
Batch size	128	32
Loss	MSE	MSE
Seeds	42	11, 23, 42, 67, 89

6.1. Mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|. \quad (2)$$

MAE remains in W/m^2 and gives every absolute deviation linear weight. It is easy to interpret but does not emphasize rare large errors.

6.2. Root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2}. \quad (3)$$

RMSE is also expressed in W/m^2 and penalizes large deviations more strongly than MAE. Its value can therefore be dominated by a small number of extreme forecast errors.

6.3. Normalized root mean squared error

$$\text{nRMSE}(\%) = 100 \frac{\text{RMSE}}{\bar{y}}. \quad (4)$$

nRMSE makes errors more comparable across locations with different mean GHI, but becomes unstable when the denominator is close to zero. It is therefore not computed on nighttime-only subsets.

6.4. Coefficient of determination

$$R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (5)$$

R^2 measures tracked variation relative to a constant mean predictor. It can be negative and should not replace scale-dependent error metrics.

The choice of a primary metric will be fixed for each resolution before the final test comparison. MAE, RMSE, nRMSE, and R^2 provide complementary evidence and will not be selectively reported according to which favors the proposed model [10].

7. Results and Discussion

7.1. Evidence from the hourly source study

Under the existing 72-hour hourly protocol, TimesNet achieved a macro-RMSE of 101.04 W/m², compared with 104.68 W/m² for the LSTM, a relative reduction of 3.48%. TimesNet also obtained lower RMSE than the LSTM at the 24- and 48-hour prefixes and lower RMSE than both temporal baselines at all three prefixes. These values describe the hourly source experiment only.

7.2. Evidence from the monthly source study

In the existing one-month-ahead experiment, the consolidated DilatedRNN forecast achieved a macro-MAE of 13.42 W/m², an RMSE of 16.20 W/m², and an R^2 of 0.898. It ranked first among five evaluated methods by macro-MAE, but led at only four of the ten locations. Local MAE ranged from 7.45 to 20.92 W/m². These values describe the monthly source experiment only and are not numerically compared with the hourly TimesNet errors.

7.3. Matched multi-resolution comparison

[Draft note: Insert the daily and matched TimesNet–DilatedRNN tables generated from the new experiment artifacts. Report one overall table by resolution and horizon, followed by location-level comparisons.]

The final analysis will distinguish three questions: which model minimizes the declared primary metric on average, how often that model wins at matched test origins, and whether its advantage is consistent across locations. A method will not be described as universally superior when the location-level evidence shows losses.

7.4. Favorable and unfavorable cases

[Draft note: Create one table for the strongest relative gains and another for the largest deficits of the proposed method. Include location, date or origin, horizon, reference GHI, both forecasts, absolute errors, and confirmed weather context when available.]

7.5. Geographic and climatic interpretation

Climate zones and factory locations will be used as descriptive context. With only ten locations, apparent differences between zones cannot by themselves establish a causal climatic effect. Sites in the same country may remain far apart and occupy distinct regimes, while nearby sites may share correlated weather. The discussion will therefore combine the map, zone classification, and error profiles without overstating statistical evidence.

8. Threats to Validity and Limitations

The NSRDB values are satellite-derived, model-based estimates and not sensors installed at the factories. The factory sample is geographic and does not represent industrial energy consumption or photovoltaic output. Monthly training series contain only four complete pre-test target years, whereas overlapping hourly forecast windows produce dependent errors. These properties limit conventional significance tests that assume independent samples.

Architecture comparisons may also be confounded by parameter count, context length, and optimization budget. The matched experiments will therefore report capacity, search space, selected configuration, random seeds, and forecast strategy. Generalization beyond the ten locations and the 2019–2024 period requires a prospective evaluation on new sites or years.

9. Conclusion

This journal extension establishes a common framework for evaluating TimesNet and DilatedRNN across hourly, daily, and longer planning resolutions at ten electric-vehicle manufacturing locations. The existing source

studies provide promising but resolution-specific evidence: TimesNet is competitive in direct hourly forecasting, while DilatedRNN is competitive for one-month-ahead forecasting from short monthly series.

[Draft note: Replace this draft conclusion after the daily and matched-horizon experiments. The final claims must follow the complete aggregate and location-level results, including unfavorable cases.]

Appendix A. Complete Hyperparameters and Supplementary Results

The final appendix will report the complete search spaces, selected hyperparameters by resolution and seed, software versions, random seeds, location-level metric tables, and supplementary forecasts. Values will be imported from retained experiment artifacts rather than transcribed manually.

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